

Recent developments of artificial intelligence methods for sea ice concentration monitoring using high-resolution imaging datasets

Marzuraikah Mohd Stofa^a, Siti Raihanah Abdani^b, Asraf Mohamed Moubark^a,
Muhammad Ammirul Atiqi Mohd Zainuri^a, Ahmad Asrul Ibrahim^a,
Nor Azwan Mohamed Kamari^a, Mohd Asyraf Zulkifley^{a,*}

^a Department of Electrical, Electronic and Systems Engineering, Universiti Kebangsaan Malaysia, Bangi 43600, Selangor, Malaysia

^b School of Computing Sciences, College of Computing, Informatics and Mathematics, Universiti Teknologi MARA, Shah Alam 40450, Malaysia

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ABSTRACT

The negative effect of sea ice shrinkage on the world's climate has been widely studied. The shrinkage of sea ice affects the heat transfer balance between the ocean and atmosphere, especially in high-latitude regions. Given the influence of global warming, the number of sea ice areas has considerably decreased over the previous decades. Hence, precise, dependable measurement of sea ice concentration is necessary to ensure the security and effectiveness of marine operations in polar areas. Monitoring of sea ice concentration through remote sensing technology is critical in predicting sea ice-related hazards, facilitating safe ship navigation and enabling polar environmental research. Furthermore, a few imaging technologies, such as synthetic aperture radar (SAR), have become the predominant tools utilised for regional ice observation because of their capability to deliver high-resolution images all year round in nearly all locations so that ice charts can be mapped effectively by national ice organisations. This study examines the utilisation of artificial intelligence (AI), particularly deep learning, in monitoring sea ice concentration through high-resolution imaging datasets obtained from several resources. These datasets include comprehensive spatial information, which is essential for accurate sea ice mapping and categorisation. Some of them focus on several main obstacles, including speckle noise in SAR images, data sparsity and amalgamation of multi-modal datasets whilst suggesting AI-based ways to mitigate these problems. This study concentrates on high-resolution imaging datasets, but it also recognises the potential use of AI in other remote sensing modalities, including scatterometry, altimetry and passive microwave sensors. This review explicitly examines research articles published from 2019 to the present, particularly those that emphasise deep learning methods. It also investigates readily available sea ice datasets, such as SAR-, optical- and drone-based datasets. Subsequently, relevant sea ice mapping techniques, including conventional algorithms, machine learning-based approaches and deep learning-based methods, are assessed. This work focuses on three main applications, namely, weather prediction, marine security and environmental preservation, and their future opportunities. Comprehensive discussions of the aforementioned topics, including the main findings and patterns, are provided to stimulate future research.

1. Introduction

Over the past few decades, high-altitude regions have been severely affected by global warming. Furthermore, the Arctic Ocean has experienced a considerable reduction in sea ice cover after the mid-1990s (Johannessen et al., 1999). Despite the observed decrease in Arctic ice's extent and thickness, the dangers and perils associated with navigating Arctic regions at present are more challenging than those in

previous periods. The Arctic is experiencing a high rate of climate change, where the increase in average surface temperatures is higher than that in other regions. Specifically, the shrinkage in Arctic Sea ice extent, when assessed at the end of the summer melting period, is rapid. The rapidly changing Arctic environment is a clear indication of climate change and thus plays an important role in influencing the global climate system. The presence of sea ice in the Arctic has restricted commercial maritime operations. Nevertheless, the frequency of

* Corresponding author.

E-mail address: asyraf.zulkifley@ukm.edu.my (M.A. Zulkifley).

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maritime activities in Arctic regions has tripled in recent years because of the prolonged duration of the open-water season (Barnhart et al., 2016). As the process of ice melting occurs, ice packs experience increased mobility, hence increasing the risks of ice floe detachment. Ice floes exhibit rapid movement patterns and could inflict harm on maritime vessels and artificial constructions (Barber et al., 2018). The monitoring of sea ice can be classified into various categories, including ice classification and detection, retrieval of ice concentration and thickness, retrieval of ice drift and detection of ice melt. Detecting sea ice is a fundamental and crucial undertaking in sea ice mapping because it enables the differentiation between areas of sea ice and open ocean.

Remote sensing, through satellite and aerial imaging, provides rapid, large-scale data collection, thus enabling frequent and timely updates. In general, remote sensing refers to the scientific and technical methodologies employed to gather data on an ecosystem process, land area, object, or phenomenon of interest through a device that does not physically interact with the investigated entity, location, or object. This information can be coupled with artificial intelligence (AI) technology to produce an automated system to monitor the environment from a distance. Recent breakthroughs in AI have considerably improved the capability to observe and classify sea ice through satellite images. AI techniques can enhance precision and automate intricate processes in remote sensing applications. AI-based solutions have garnered attention because of their capacity to analyse extensive information and generate intricate sea ice maps with limited human involvement. In general, the advancement of remote sensing technology has substantially contributed to many applications in various domains, including forest segmentation (Ru et al., 2023), environmental monitoring (Kabir et al., 2022), seashore safety (Stofa et al., 2020) and agriculture automation (Zhu et al., 2020a).

Conventional techniques for identifying sea ice through remote sensing imaging involve the utilisation of pixel-based algorithms, which rely solely on the spectral profile of each pixel to extract sea ice masks (Hossain and Chen, 2019). For example, in the work of Sandru et al. (2020), shipborne cameras were used to capture data, which were then analyzed using conventional computer vision methods, including spatial filtering, k-means clustering and thresholding, to identify different types of sea ice categories and concentrations. The application of conventional computer vision methods in real-time object detection is inefficient in scenarios where environmental changes are unavoidable. This issue is important for any intelligent system, given that challenging conditions are prevalent in the polar sea, which is characterised by diverse lighting circumstances, including fluctuating illumination levels, glare from sunlight interacting with ice and water surfaces, reflections on water and various ice formations. Even though conventional methods have produced promising outcomes in sea ice segmentation, they cannot generate high-precision maps when they are applied to highly detailed, high-volume remote sensing images, especially those from the most recent satellites.

Recent developments in deep learning-based AI technology have become essential tools for resolving these difficulties. Utilising AI-driven techniques, researchers have produced accurate sea ice maps, thereby facilitating the precise monitoring of sea ice dynamics and their consequences on climate change. The use of advanced AI techniques in satellite imagery successfully improves the precision and efficiency of intelligent systems that have been proven useful in diverse application domains, including stock market detectors (Zulkifley et al., 2023), cyber security (Zhang et al., 2022b), agriculture systems (Zulkifley et al., 2022), urban planning (Sanchez et al., 2023) and deforestation detection (Md Jelas et al., 2024). These advanced AI techniques, also known as deep machine learning, enable the extraction of insightful information and features from vast amounts of satellite data, which are then used to formulate well-informed decisions. The application of deep learning techniques in sea ice classification has automated the laborious, time-consuming process that would otherwise require manual annotation and analysis. Moreover, it facilitates the continual surveillance of

sea ice conditions because it is an automated process, thus enhancing human knowledge on any ecological transformation in polar areas. Given that advanced AI techniques have contributed substantially to sea ice mapping based on remote sensing data, many deep learning techniques, particularly the convolutional neural network (CNN), have been designed to improve the accuracy of interpreting remote sensing images, especially for the segmentation of sea ice regions. Malmgren-Hansen et al. (2021) experimented on various CNN models in a situation with two different resolution data between Sentinel-1 synthetic aperture radar (SAR) and AMSR 2 sensors. Their study concluded that CNN is well-suited for combining data from several sensors with a high level of reliability. Furthermore, DenseNet (Huang et al., 2017) has shown outstanding performance in extracting sea ice concentration from SAR images, with training and testing errors of only 5.24 % and 7.87 %, respectively. In consideration of these development patterns in the past decade, various studies related to sea ice, especially sea ice mapping, have been gaining attention in line with the technological advancements led by the introduction of deep learning techniques. Fig. 1 shows the rate of increment in the number of research publications on sea ice-related topics from 2019 to 2023. The trend demonstrates the increasing interest in applying deep learning methodologies in sea ice monitoring, motivated by the need for precise climate forecasts and marine safety. Furthermore, the increasing trend in publications underscores the importance of AI developments in this field. (See Table 1.)

Sea ice concentration is a vital parameter for comprehending the dynamics of sea ice, which directly influences global climate systems and marine activities. Monitoring sea ice concentration provides vital data on the quantity of sea ice coverage and its seasonal fluctuations, which are crucial for climate modelling and navigation safety in polar areas. This review examines the application of AI, particularly deep learning, in the monitoring sea ice concentration by using high-resolution imaging datasets. These datasets are derived from a few imaging sensors, including SAR, optical and drone-based sensors, which provide detailed spatial resolution that is necessary for accurate sea ice mapping. In contrast to the review of Lyu et al. (2022), which extensively addressed SAR-based methodologies for sea ice detection, our review focuses on the integration of sophisticated AI approaches with SAR, optical and drone datasets for automated and scalable sea ice monitoring. Scatterometry, altimetry and passive microwave sensors, although valuable for large-scale monitoring, typically provide low spatial resolution or non-imaging data, making them incompatible with this review. By narrowing the focus to high-resolution imaging datasets,

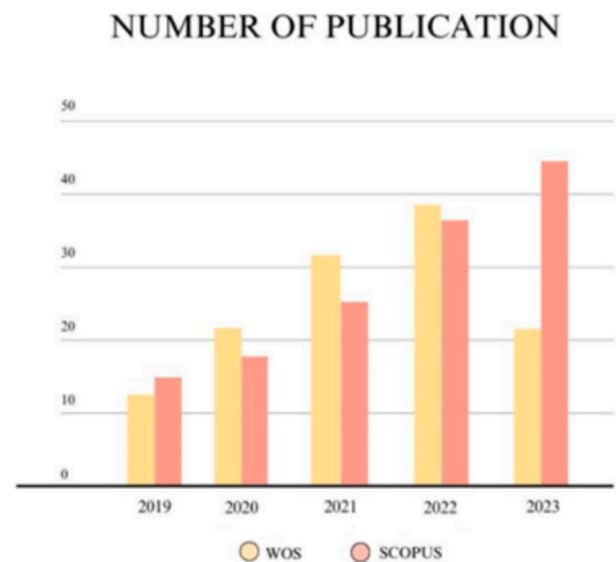


Fig. 1. Statistics of the published research papers on sea ice segmentation according to the WOS and SCOPUS databases.

Table 1

Sea ice classes with their respective descriptions.

Class	Description
Pancake ice	Predominantly circular ice fragments ranging in diameter from 30 cm to 3 m and reaching a maximum thickness of around 10 cm, featuring elevated borders resulting from the impact of the pieces with each other.
Level ice	Sea ice that remains undamaged over time.
Ice floe	Any sea ice that is contiguous.
Iceberg	An iceberg fragment that is currently adrift at sea.
Floe bit	A stretch of sea generally no wider than 10 m, consisting of an icy hummock (or multiple ridges) isolated from the surrounding environment. It typically rises to a height of 2 m above sea level.
Floe berg	A substantial expanse of sea ice consisting of a solitary hummock or a collection of hummocks that have solidified and are isolated from the surrounding ice. It typically rises to a height of 5 m above sea level.
Deformed ice	A generic designation for ice that has undergone compression and, at certain locations, upward and downward forces. Ice is subdivided into rafted, ridged, and hummocked sections.
Brash ice	Floatable ice accumulations consisting of fragments no larger than 2 m in diameter, as well as the remnants of other types of ice.
Broken ice	Predominately flat ice cover that has been fractured by gravitational waves or melting decay.
Underwater ice	The portion of a sea ice object that is visible but submerged.
Melt pond	Lakes that develop on the surface of the ocean throughout the spring and summer
Open water	A substantial expanse of open-to-travel water where the concentration of sea ice is below 1/10.

this work aims to provide a comprehensive overview of these datasets' applications and challenges in AI-driven sea ice concentration monitoring. Then, previous studies have explored various remote sensing techniques for sea ice monitoring; for example, [Lewis et al. \(2019\)](#) demonstrate the feasibility of detecting sea ice using single-channel shortwave infrared data. However, this study is often constrained by a single sensor modality that limits their real-time and large-scale monitoring scalability. Additionally, the lack of standardised benchmarking frameworks makes comparing AI model performance across different datasets and methodologies challenging.

This review addresses these gaps by emphasising the integration of multi-source high-resolution imaging datasets, such as SAR, optical, and drone-based sensors, to enhance the scalability and accuracy of AI-driven sea ice concentration mapping. Then, this review thoroughly examines current developments in sea ice concentration mapping technology, emphasising the utilisation of deep learning-based AI technologies over the last five years. Whilst previous studies focused on conventional machine learning methods and general remote sensing techniques, this study thoroughly assesses the aforementioned deep learning methods, including their merits and demerits. It advocates for additional investigation into the optimisation of sea ice monitoring in diverse applications. In consideration of the previous contexts, the following shows the primary contributions of our work.

1. This detailed review of sea ice mapping uses AI based on satellite images.
2. This review comprehensively explains sea ice categories and open-access databases.
3. This article describes in depth how conventional and deep learning techniques are utilised in sea ice concentration mapping.

The novelty of this study lies in its detailed study of deep learning methodologies for monitoring sea ice concentration through high-resolution imaging datasets. This study enhances the field by integrating multi-source remote sensing data and establishing a standardised benchmarking framework, tackling key challenges in sea ice segmentation and classification. The implementation of deep learning techniques has elicited considerable interest from international

researchers.

The subsequent sections of this article are structured as follows. [Section 2](#) presents the research methodology, and [Section 3](#) provides a comprehensive description of sea ice features and types. The details of several main open-access sea ice datasets are described in [Section 4](#). [Section 2](#) reviews traditional and conventional machine learning methods that rely on heuristic algorithms, statistical models, or simple machine learning approaches. These methods typically depend on pre-defined rules and expert-defined thresholds. [Section 5](#) focuses on advanced deep/machine learning techniques that utilise large-scale data and complex network architecture to achieve state-of-the-art results for sea ice concentration monitoring systems. General discussion remarks on sea ice mapping are presented in [Section 6](#). The conclusions, challenges and future research opportunities in the field of sea ice mapping are summarized in [Section 7](#).

2. Research methodology

The development of this review follows a structured four-step methodology to ensure clarity and thoroughness in analysing AI-based sea ice concentration monitoring techniques.

The first step defines the research objectives by identifying key research problems related to sea ice concentration monitoring. It establishes a focused scope by emphasising AI-based methods and high-resolution imaging datasets, including SAR, optical, and drone imagery.

The second step involves the selection and categorisation of studies. Relevant literature from peer-reviewed sources is examined, with a particular focus on identifying sea ice features and types. Additionally, open-access sea ice datasets are analyzed to assess their suitability for deep learning applications.

The third step presents a comparative analysis of AI-based techniques. The selected methodologies are categorized into conventional machine learning and deep learning approaches. This stage also involves a detailed comparison of their strengths, limitations, and effectiveness in sea ice mapping. Trends and key findings in AI-driven sea ice monitoring are highlighted to provide a comprehensive understanding of recent advancements.

The final step discusses the implications and conclusions drawn from the review. AI advancements in sea ice concentration monitoring are summarized, along with the challenges encountered, such as speckle noise and dataset limitations. The discussion also identifies potential future research directions to improve the accuracy and efficiency of AI-based sea ice monitoring systems. [Fig. 2](#) illustrates the comprehensive information that outlines the working flow of this article.

3. Sea ice features and types

Sea ice is a common natural phenomenon on Earth. It is prevalent in regions with high latitudes and spans around 25 million km² of the earth's ocean surface, equivalent to one-tenth of the total ocean area. Sea ice is a crucial factor that influences the balance of climate systems, heat transfer stability, the ocean water cycle and atmospheric circulation ([Park et al., 2020a](#); [Sandven et al., 2000](#)). Table I summarises the descriptions of sea ice classes, and [Fig. 3](#) shows the stages of sea ice development. Comprehending these categories is essential for designing deep learning models that can effectively classify and segment diverse sea ice classes by using remote sensing data. The classification of various sea ice categories influences the efficacy of deep models in identifying perilous navigation circumstances. (See [Fig. 4](#).)

Sea ice concentration denotes the ratio or proportion of a particular region that is enveloped by sea ice. It is an essential metric used to quantify the geographic range of sea ice in a specific area, which is vital for comprehending the distribution of sea ice coverage. Sea ice concentration is quantified as a percentage, where 0 % represents the absence of ice (open water) and 100 % represents the presence of ice covering the entire area. Sensing algorithms utilise passive microwave

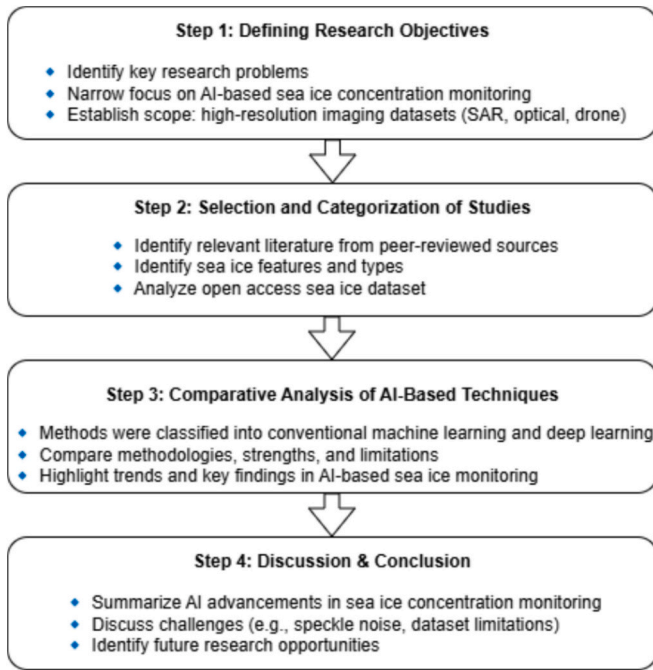


Fig. 2. Overall information flow of the proposed review work.

data from many channels with varying frequencies or polarisations to capture sea ice features. Over time, numerous algorithms have been developed, with the majority of them relying on data obtained from 19 and 37 GHz channels (Laverne et al., 2019). Many sea ice concentration algorithms rely on tie points that characterise the usual radiative characteristics of ice and water. When the channels utilised by these algorithms are defined, the tie points are specified as luminance temperatures or emissivities, and these points vary for each surface condition. These parameters cause variations across algorithms, making it difficult to guarantee that each algorithm will produce a sea ice concentration of 100 % for consolidated ice areas and 0 % for open-water regions. Fig. 4 shows a sample of sea ice concentration illustration from Hudson Bay region.

4. Open access sea ice dataset

Large amounts of data are crucial in fitting a deep learning model. These data are collected into a dedicated dataset that is usually accompanied with ground truth labels. In the era of deep learning, many datasets are easily accessible, especially for well-established challenges, such as object identification, pedestrian tracking and autonomous driving. Typically, a dataset needs to be of considerable size, which can also be augmented through synthetic data. Lack of training data poses an obstacle in fitting the deep learning model optimally, especially in semantic segmentation and classification of sea ice. In reality, acquiring adequate sea ice data with verified labelling can be challenging, and manually annotating each datum at a pixel level can be an excessively time-consuming process.

To ensure an accurate annotation process, ice service specialists utilise various data sources, including in-situ visual identification, satellite imagery, ice radar and any available satellite, to detect and classify sea ice in its targeted operational environments. One of the important sets of annotated data on sea ice was produced by the International Maritime Organisation, which established a direct correlation between the operational risk of a vessel and the concentration of various sea ice varieties in a given area. Presently, multiple sea ice charts, such as those provided by the Russian Arctic and Antarctic Research Institute, the U.S. National Ice Center and the Canadian Ice Service (CIS), can be accessed for research purposes. This research employs multiple essential datasets to assess deep learning methodologies for monitoring sea ice. The reviewed datasets include CIS ice charts, AI4SeaIce and SAR-based datasets, which are sourced from the Sentinel-1 satellite. These datasets are selected because of their extensive coverage of various sources and their pertinence to sea ice mapping activities. CIS ice charts also contain expert-validated annotations, making them a dependable ground truth reference. The AI4SeaIce dataset comprises high-resolution SAR images accompanied with their respective ice charts, thereby facilitating the efficient training and assessment of automated segmentation algorithms.

This review focuses on frequently employed CIS-provided ice charts that serve as the baseline dataset. CIS ice charts provide vital information with extensive coverage and precision, and they have been validated using several data sources, including satellite imaging, in-situ observations and ice radar. The essential criteria for developing a

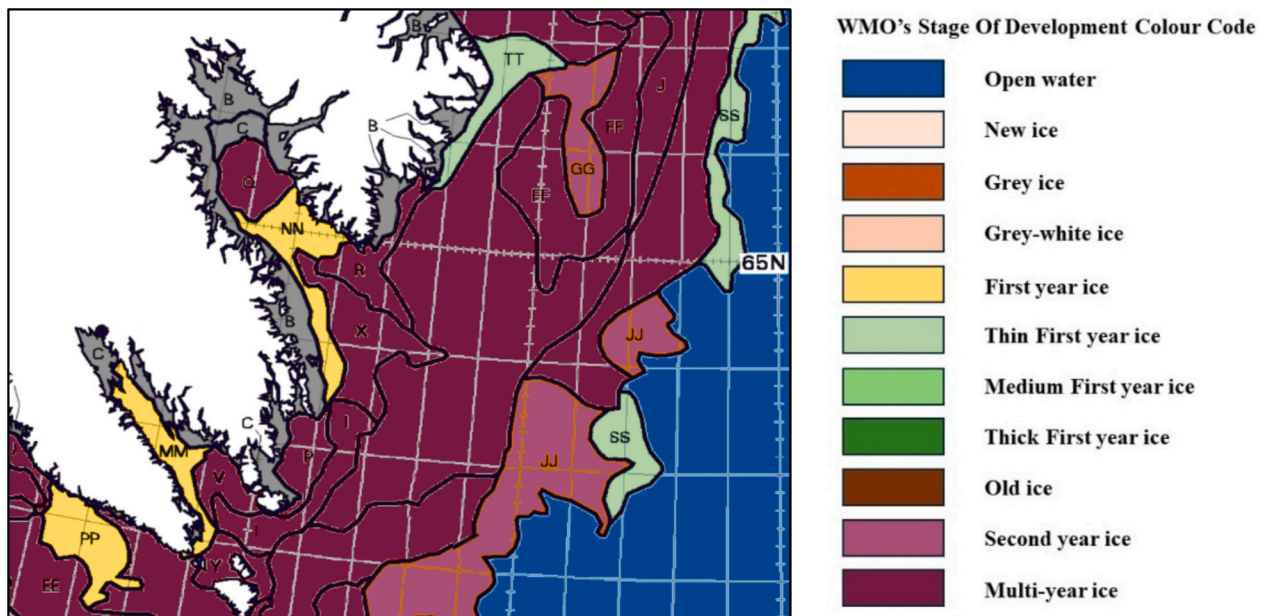


Fig. 3. Stages of sea ice development.

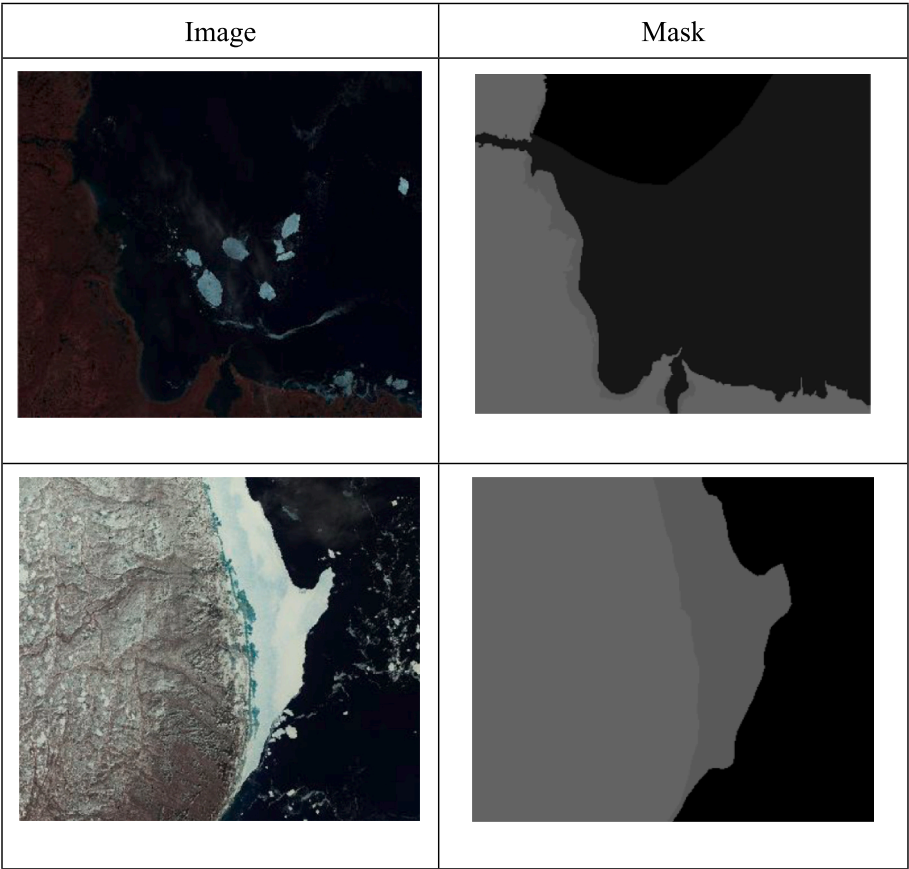


Fig. 4. A sample of sea ice concentration illustration.

benchmark for sea ice concentration investigations to produce accurate ground truth rely on expert assessment and cross-referencing with high-resolution satellite data and in-situ observations. The precision and dependability of deep learning models for monitoring sea ice concentration rely on a set of comprehensive benchmarking datasets. This research predominantly employs high-resolution imaging datasets, including SAR, optical and drone-based imagery, to evaluate the state-of-the-art AI model's performance. CIS ice charts function as the principal benchmarking dataset because of their expert-validated annotations that integrate many sensors coupled with elevated geographic information. The primary benchmarking criteria employed in this study are defined below.

- 1. Resolution and data quality
 - The datasets employed for benchmarking must possess high spatial resolution to facilitate accurate sea ice concentration mapping.
 - SAR imagery (e.g. Sentinel-1) and optical imagery (e.g. MODIS, Landsat and drone based) must provide comprehensive data that facilitate deep learning-based segmentation.
- 2. Consistency and temporal perspective
 - CIS ice charts offer long-term and multi-seasonal data that facilitate benchmarking across various ice conditions and periods.
 - AI models developed using high-resolution datasets are assessed against multi-year records to verify generalization across seasonal fluctuations.
- 3. Evaluation of performance metrics
 - Benchmarking of AI models requires a set of standardised quantitative measurements, such as the Dice coefficient, intersection over union (IoU), pixel-wise accuracy, precision and recall.

- These metrics establish a standardised framework for assessing deep learning models in relation to high-resolution imaging benchmarks.

CIS ice charts are the primary benchmarking dataset for assessing deep learning models in sea ice concentration research. The charts are manually annotated by professionals and validated by various remote sensing sources, including SAR data (Sentinel-1), optical imagery (MODIS and Landsat) and in-situ measurements. Evaluating AI models by using CIS ice charts adheres to the following structured process.

- 1. Multi-source validation ensures that CIS ice charts accurately represent sea ice conditions, providing a highly dependable ground truth dataset.
- 2. Temporal consistency enables the evaluation of AI models over diverse seasonal variations, hence ensuring the robustness of predictions.
- 3. The geographic breadth facilitates benchmarking across extensive Arctic areas, rendering CIS ice charts flexible for varying ice conditions.
- 4. Standardised performance validation criteria, such as the Dice coefficient, IoU and precision-recall scores, are employed to evaluate AI model outputs against sea ice concentration data collected from CIS.

These well-designed datasets are essential for various applications, including sea navigation, environmental monitoring and climate modelling. Hence, a solid framework must be established to assess and advance deep learning applications in sea ice concentration research. At present, the proliferation of openly accessible datasets pertaining to sea ice is commensurate with the increasing interest of researchers in this

topic. This review compiles an exhaustive set of datasets with the intention of fulfilling the basic research requirements for experimental evaluations and establishing a standardised framework for future benchmarking investigations. This review focuses on high-resolution imaging datasets, including SAR, optical and drone-based data, for monitoring sea ice concentration. These datasets are widely used because of their detailed spatial resolution and compatibility with AI-driven methodologies. In addition to comprehensive source descriptions, eight datasets are gathered, as shown in Table 2.

1. Optical sensor datasets (Kang et al., 2022)

The dataset is composed of 50 m resolution visible-light images from HY-1. Geographically, the images encompass the Bohai Sea region in China. The given images have diverse dimensions ranging from 512 to 2048 pixels and comprise more than 2500 images. Sea ice has been diligently annotated for each pixel in each image, resulting in the separation of the data into two classes: background and sea ice. The RGB channels are included in the TIFF format of the remote sensing images, whereas the annotation files have a single channel and are saved in PNG format. The annotation files allocate a value of 255 to sea ice pixels and a value of 0 to background pixels.

2. SAR dataset

SAR, which has been applied in 80 % of sea ice concentration publications, is the most frequently utilised active microwave data type. This study compiles five open datasets as follows:

Table 2
The overview of the detailed description of the eight datasets.

Dataset	Type	Area	Source
2021Gaofen (Kang et al., 2022)	Optical	Bering Strait, China	HY-1 visible light imagery
Arctic Sea Ice Cover Data From Spaceborne Synthetic Aperture Radar By Deep Learning (Wang and Li, 2021)	SAR	Artic	Sentinel-1 dual-polarisation data in extra wide swath (EW) mode.
AI4SeaIce (Stokholm et al., 2022)		Baffin Bay, Western Greenland	Sentinel-1 dual-polarized
Physics-Aware Feature Learning Of Sar Images With Deep Neural Networks (Huang et al., 2021b)		Belgica Bank	Sentinel-1 Interferometric Wide (IW)
SI-STARS-7 (Song et al., 2021)		Hudson Bay	Sentinel-1 dual-polarisation data in extra wide swath (EW) mode.
NWPU_YRCC dataset (Zhang et al., 2020)		Ningxia-Inner Mongolia	Drone Canon 5DS visible light camera and a DJI Inspire 1
Sea Ice Classification via Deep Neural Network Semantic Segmentation (Dowden et al., 2021)	Alternative techniques of acquiring datasets	Ross Sea in Antarctica	Go-pro images captured by Nathaniel B. Palmer
Global Navigation Satellite System (GNSS) signals (Yan et al., 2019)		Canada	(GNSS) signals

a) Arctic Sea Ice Cover Data From Spaceborne Synthetic Aperture Radar By Deep Learning (Wang and Li, 2021)

The dataset comprises coverage data of Arctic sea ice and is derived from Sentinel-1 SAR. The Arctic region has access to an estimated 2500 SAR sequences per month. Every Sentinel-1 SAR image obtained in the Arctic has undergone processing to produce NetCDF data representing the extent of sea ice coverage. Every Sentinel-1 image correlates to a corresponding NC file. Sea ice cover derived from SAR has a 400 m spatial resolution. The website has made available the processing of Sentinel-1 data collected in the Arctic between 2019 and 2021, together with the related sea ice coverage statistics.

b) AI4SeaIce (Stokholm et al., 2022)

The dataset comprises 461 Sentinel-1 SAR sequences compiled between 2018 and 2019 that have been matched with ice charts generated by the Danish Meteorological Institute. The ice charts present data on sea ice concentration, stage of ice development and shape of the ice in the form of manually created polygons. AMSR2 microwave radiometer sensor measurements are also included in the dataset to aid in the training of sea ice concentration; however, their resolution is much lower than that of Sentinel-1 data. Song et al. (2022) transformed the AI4SeaIce dataset into a dataset dedicated to the semantic segmentation of ice water.

c) Physics-Aware Feature Learning Of Sar Images With Deep Neural Networks (Huang et al., 2021b)

The objective of this dataset is to investigate the classification of sea ice by using Sentinel-1 Interferometric Wide SAR images, along with Ground Range Detected High-resolution (GRDH) and Single-look Complex products, into the HH channel. The area of interest chosen is Belgica Bank, a region covered in ice located along the northeast coast of Greenland at approximately 79°N and 16°E. The GRDH image is divided into non-overlapping patches measuring 256 × 256 pixels. Each patch is labelled with eight semantic labels that primarily pertain to sea ice.

d) SI-STARS-7 (Song et al., 2021)

The dataset consists of SAR imagery specifically curated for sea ice classification, focusing on spatial and temporal aspects. The SAR data were collected during two Hudson Bay freeze-up periods: between October 2019 and May 2020 and between October 2020 and April 2021. The dataset comprises an extensive variety of ice classifications. The designations for various types of sea ice are obtained from CIS' weekly regional ice charts. Each data sample comprises dual-polarisation (HH and HV) SAR data and represents a 32 × 32 pixel region of SAR imagery. The patches are obtained from a series of six successive SAR scenes, adding a time aspect to the dataset. This dataset contains a total of 80 SAR scenes to produce the necessary data. The study encompasses all categories of sea ice found in Hudson Bay, which consist of seven distinct types: thin first-year ice, medium first-year ice, thick first-year ice, open water, new ice, grey ice and grey-white ice.

3. Alternative techniques of acquiring datasets

Drone- and icebreaker-captured images are examples of alternate acquisition methods used in ice classification datasets. A total of 305 ideal images are chosen from videos and images obtained through the use of drones.

1. NWPU_YRCC dataset (Zhang et al., 2020)

This dataset was obtained annually along the Ningxia-Inner Mongolia section of the Yellow River between November and March

from 2015 to 2019. The images were trimmed to 1600×640 pixels and contain four target classes. Most of these photos were taken during the freezing phase. The Photoshop software was utilised to classify every pixel in the images into one of four distinct categories: water, migrating ice, coastal ice or others. At a 6:2:2 ratio, the dataset was divided into training, validation and testing sets, each containing 61, 183 and 61 images, respectively.

2. Camera-based airborne datasets (Dowden et al., 2021)

The dataset was compiled from GoPro images captured by Nathaniel B. Palmer during his two-month journey across the Ross Sea in Antarctica. The images that comprise the dataset vary slightly in quality (ranging from HD to 4 K HD) and consist of images taken from a fixed location on an icebreaker. The images were collected at regular intervals throughout each day, with a total of 1600 images captured per day. Approximately 60 images were selected in a random manner to be labelled for each day in the dataset. Each image was categorized into eight distinct classifications in accordance with the pixel-wise labels that correspond to its components. The pixel-wise labels were generated through meticulous and comprehensive annotation by using the PixelAnnotationTool [32], which employs the watershed labelling technique. The dataset has eight classes; four are allocated to describing non-ice objects, and four are allocated to ice types. The following text sums up non-ice objects.

- Lens: any particulate or object that obstructs the image and presents on the camera lens
- Sky: the entirety of the visible sky within the image
- Vessel: all visible portions of the boat from the front, including flag, crew and mast
- Ocean: the image contains only open water

3. Global Navigation Satellite System (GNSS) signals (Yan and Huang, 2019)

GNSS reflectometry (GNSS-R) has been used as an innovative tool for sea ice monitoring systems. Datasets, such as TechDemoSat-1 (TDS-1), employ reflected GNSS signals to deliver data on sea ice extent, roughness and concentration. These datasets demonstrate the potential to enhance classic SAR and passive microwave datasets by providing distinct insights into ice surface features. Recent research in the Canadian region that used GNSS-R data for monitoring Arctic sea ice found that such data are effective in dynamic environmental settings. Despite the underutilisation of GNSS-R datasets in sea ice research, their integration with current remote sensing techniques offers a viable research direction for future investigations.

5. Conventional sea ice concentration mapping methods

Sea ice mapping is the identification and delineation of sea ice regions on aerial or satellite imagery. Monitoring the changes in sea ice extent is crucial because of its massive implications for climate research and environmental studies. Consistent with these aims, the subsequent subsections elaborate on the diverse conventional sea ice mapping techniques employed for the segmentation of sea ice regions.

Despite the increased prevalence of deep learning approaches in recent years, conventional approaches are still valuable, especially in cases with small amounts of training data, which hinder the optimal fitting of a deep learning model. Conventional approaches are ideal for semantic segmentation under specific circumstances. These methods concentrate on binary segmentation tasks, including watershed segmentation, thresholding and active counting. These techniques have demonstrated their effectiveness in accurately retrieving sea ice characteristics from digital images for good-quality imagery input. The adaptive thresholding technique, as demonstrated by Jumb et al. (2014),

offers a means of segmenting colour images on the basis of a dynamically chosen threshold. Another prevalent method utilised for conventional semantic segmentation is texture-based segmentation. For example, Deng and Manjunath (2001) introduced an unsupervised method for segmenting regions on the basis of colours and textures. An algorithm that integrates adaptive thresholding and texture-sensitive super-pixeling was introduced by Chai et al. (2021) to improve the precision and robustness of the segmentation outputs. Achanta et al. (2008) presented a segmentation technique for detecting salient regions by utilising low-level characteristics of colours and luminance, which are then used to segment the image accordingly. Meanwhile, Liu et al. (2016) adopted a hybrid approach of active contour and curvelet transform and developed an algorithm for automated detection of sea ice edges by using SAR image input. A gradient vector flow snake technique was introduced by Zhang et al. (2015) to excerpt the boundaries of ice floes. This approach can autonomously detect and differentiate inadequate boundaries amongst floes. By effectively differentiating ostensibly interconnected ice floes into discrete components, the algorithm demonstrates strong segmentation performance. The graph cuts method proposed by Boykov and Jolly (2001) is another conventional approach to object segmentation. This method is an interactive approach to segmentation, where the user provides early markers to create the boundaries that surround the chosen objects. Generating effective segmentation maps can be a challenging task because of the irregular illumination conditions that occur during the day and the combined glare effect from ice and ocean surfaces (Geffray et al., 2016). The diversity of sea ice characteristics, including type, size, colour, texture and shape, creates extra challenges that render conventional image processing methods minimally effective (Wright and Polashenski, 2018). This issue is increasingly complicated by the diverse environmental conditions of Arctic regions that face frequent fog, rain, limited sunlight and heavy snow. As a result, conventional segmentation solutions, with their limited learning capability to discern variations in ice colour, shape and texture, may become unreliable and require fine-tuning. In general, conventional image segmentation methods can perform basic segmentation tasks with high efficiency. However, as the input images become increasingly complex, ascertaining the optimal threshold values to accurately segment multiclass objects becomes difficult. Moreover, the luminance and noise levels in input images can substantially affect threshold selection, which restricts the capability of conventional methods to extrapolate the correct boundaries for different contexts.

Since the inception of machine learning techniques, numerous classical machine learning models have been applied to sea ice image segmentation to compensate for the weakness of conventional segmentation methods. Support vector machine (SVM) and random forest (RF), which are traditional machine learning techniques, exemplify early developments in AI for sea ice mapping. These strategies, which focus on manually derived features, have proven to be essential in the advancement of automated sea ice detection methods. Machine learning models have advantages over standard linear segmentation algorithms because of their capability to project data samples into a space with a high dimensionality. This capability greatly facilitates the training of a precise sea ice classifier. A novel approach for distinguishing amongst distinct classes of sea ice types by using SAR data was developed by Shu et al. (2020); the approach is based on the object-based RF model. This classifier exhibits exceptional performance in accurately classifying various characteristics of Arctic sea ice and effectively mitigating the issue of misclassification in the border regions. Meanwhile, Kalke and Loewen (2018) applied an SVM model on remote-sensing image data to monitor the characteristics of river ice by training three SVM classifiers to differentiate between water and ice; the model was then used to forecast the overall surface ice concentration. Park et al. (2020b) introduced an RF-based sea ice classification algorithm. A set of SAR images of sea ice was classified using the trained classifier, and the model's ability to accurately identify daily ice patterns was validated using the accuracy metric.

The ice segmentation outcomes produced by these conventional machine learning algorithms are superior to those produced by the threshold-based approach when tested on basic-quality remote sensing imaging data. However, the training process can become exceedingly intricate when complex image sequences are involved because feature extraction must be performed manually prior to network fitting.

Polarisation features are essential for the identification and classification of sea ice because they facilitate the differentiation of various sea ice types and the comprehension of their physical characteristics. For example, the utilisation of various polarisation states in polarimetric SAR (PolSAR) enhances the differentiation of ice varieties. Classification accuracy is enhanced, particularly in complex scenarios (e.g. sea ice melting periods), through the capability to extract and analyse these features, such as co- and cross-polarisation ratios.

Polarimetric techniques have been shown to be effective in the monitoring of sea ice in recent studies. For instance, [Zhang et al., \(2024\)](#) developed the polarimetric autocorrelation matrix, a tool that improves the accuracy of sea ice classification by adopting target polarisation and Doppler scattering mechanisms. [Wang et al., \(2024\)](#) also emphasised the importance of PolSAR data during the sea ice melting period, demonstrating that multi-frequency polarimetric data can substantially enhance the performance of sea ice classification. Meanwhile, the efficacy of co- and cross-polarisation ratios from RADARSAT-2 imagery in the detection of sea ice was demonstrated by [Zhao et al., \(2024\)](#). PolSAR has been extensively employed in sea ice applications to improve detection and classification capabilities. [Shokr and Dabboor, \(2024\)](#) conducted a thorough examination of these applications, with emphasis on the enhancement of the accuracy of sea ice classification models by various polarimetric data, particularly in challenging conditions.

In summary, conventional approaches achieve high efficiency when they are applied to straightforward binary segmentation tasks. However, when the input image scenes exhibit complex characteristics, such as

heterogeneous ice types (e.g. a mixture of first-year ice, multi-year ice and melt ponds), fluctuating lighting conditions (e.g. shadows, glare, or low contrast) and elevated noise levels typically present in SAR imagery, ascertaining appropriate threshold values for precise object delineation across multiple class categories becomes progressively difficult. In addition, threshold selection is susceptible to normally occurring challenges, such as image noises and luminance variations, and this susceptibility restricts their ability to generalise when applied to various imaging scenes. Although conventional approaches have their merits, their drawbacks provide an opportunity to investigate advanced modelling that can address the aforementioned difficulties. Researchers have endeavoured to overcome these challenges in threshold-based segmentation through the utilisation of sophisticated methodologies, including machine learning, probabilistic models and adaptive algorithms. These alternative approaches present encouraging prospects for improving segmentation precision, especially in intricate scenes, and reducing the effects of luminance and noise. Although conventional sea ice mapping methods achieved remarkable progress up until 2019, a notable shift towards the integration of these methods with fresh techniques, such as AI and deep learning, has been observed in recent years. This situation has decreased the number of new studies that exclusively focus on conventional methodologies in the post-2019 period. [Table 3](#) shows an overview of conventional approaches for sea ice concentration mapping.

6. Deep learning sea ice concentration mapping methods

The deep learning field has made substantial progress, especially in image segmentation and classification, which are applied to various domains, such as sea ice concentration monitoring. CNNs effectively classify sea ice from SAR data and have higher accuracy than conventional machine learning methods. This high accuracy is achieved by

Table 3
An overview of the conventional approach for sea ice concentration mapping.

Source	Year	Input Source	Application	Method/Model	Strength	Weakness
Zhang et al. (2015)	2012	Ice tank at the Hamburg Ship Model Basin (HSVA) in the summer of 2011	Analyse an evolving broken ice	Object extraction and edge detection	The Otsu thresholding approach is highly effective for analysing images and videos of the ice tank.	Failure to accurately recognise the limits of the ice floe will lead to missing or incomplete boundary information, which in turn will have an impact on the statistical outcomes.
Liu et al. (2016)	2016	RADARSAT-2 SAR Images	Detecting the ice edge	Curvelet Transform and Active Contour	Helpful as a first guess for ice analysts or to assimilate SAR data.	The ice edge might be overestimated because of the curvelet transform's scale dependence and the active contour's internal force.
Kalke and Loewen (2018)	2018	Digital images	Monitoring river ice	SVM	The ice concentration values obtained from SVM-generated binary images were notably more accurate in comparison to thresholding approaches.	Additional investigation is required to comprehensively evaluate the capability of the SVM model in distinguishing between the frazil and anchor ice components of the total surface ice.
Shu et al. (2020)	2019	Cryosat-2 satellite data	Discriminate different sea ice types	ORF	Minimizing misclassification at the boundary between distinct ice kinds	Snow loading may affect the shape of the waveform.
Park et al. (2020b)	2020	Sentinel-1 SAR images	Sea ice type classification	Random forest	The study emphasises the possibility of providing ice mapping services in near-real-time by utilising daily mosaicked images from the Sentinel-1 satellite.	The study acknowledges that the algorithm's performance is affected by the inconsistency of ice charts and the high generalization level
Chai et al. (2021)	2021	Remote sensing images	Segmenting sea ice floes	Adaptive thresholding and texture-sensitive superpixeling	The proposed method utilizes a two-stage process of learning-based adaptive thresholding to produce a more precise segmentation.	Limit the generalizability of the results to a broader range of images or different geographical regions.
Wang et al. (2024)	2024	PolSAR data	Classifying sea ice	Random forest SVM	Enhancements in classification accuracy, with RF classifiers capable of attaining up to 95.84 %.	The limited feature set investigation and the complexity of multi-band data
Zhao et al. (2024)	2024	RADARSAT-2 Imagery	Sea ice detection	Random forest K-means SVM Decision tree Bayesian	Enhancing the precision of sea ice detection, which is essential for comprehending the effects of climate change on Arctic ecosystems and navigation	Necessitate a high level of expertise and can be time-consuming, due to the require extensive feature engineering.

utilising high-resolution satellite imagery to generate precise predictions of the extent of sea ice during critical periods. Network architectures that are based on deep learning have emerged as the standard de-factor model for recent semantic segmentation applications. Whilst SAR and optical datasets remain the dominant sources for AI-based sea ice mapping, recent studies have explored the usage of GNSS-R as an alternative data source. GNSS-R data, such as those from TDS-1 and CYGNSS, provide indirect observations of sea ice properties on the basis of reflected GNSS signals. Some studies have applied CNN models to classify sea ice concentration by using GNSS-R-derived features, demonstrating its potential as a complementary dataset. [Yan and Huang \(2018\)](#) pioneered the application of CNNs for this purpose, illustrating CNNs' ability to extract significant features from GNSS-R signals. Their research established a framework for subsequent studies, including our proposed methodology, which enhances feature extraction and classification precision. [Zhu et al. \(2020b\)](#) explored machine learning methodologies, specifically decision trees and RFs, to assess sea ice by utilising 42 months of GNSS-R data from TDS-1. They identified characteristics from 2D delay waveforms with differing Doppler spreads and utilised this information to classify sea ice occurrence. The RF-based approach obtained an overall accuracy of 98.03 % in the Arctic and 95.96 % in the Antarctic, illustrating the effectiveness of using GNSS-R data as an input to a machine learning model for sea ice monitoring purposes. [Yang et al. \(2023\)](#) presented a deep neural network methodology for extracting Arctic sea ice concentration by utilising GNSS-R data from the TDS-1 satellite. Through the examination of delay-Doppler maps and the consideration of sea ice–ocean interactions, their methodology obtained root mean square errors (RMSEs) of 0.0284 in March and 0.0415 in June 2016 with respect to the reference datasets. The integration of GNSS-R-derived sea ice concentration data into these models enhanced the precision of the Arctic sea ice forecasts, especially in marginal areas, and achieved regional RMSE reductions of up to 17 % in 24-h forecasts. However, GNSS-R's lower spatial resolution compared with that of SAR and optical imagery presents challenges in producing detailed sea ice segmentation, limiting its standalone applicability to AI-driven concentration mapping. Many network architectures have been developed to perform semantic segmentation, and each proposed model has been designed and fine-tuned for a specific application domain, featuring a unique compromise between accuracy and efficiency. The base CNN layers can autonomously learn unique characteristics from images, resulting in precise comprehension of the scenes of interest.

The subsequent paragraphs are explained as follows. The first paragraph delineates the progression of deep learning architectures for image segmentation, from fully convolutional networks (FCNs) to sophisticated models, such as SegNet, U-Net and PSPNet. The second paragraph explores the recent enhancements in model performance and the advancements and applications of CNN in the prediction and classification of sea ice concentration. The third paragraph examines developments in semantic segmentation achieved through deep learning, highlighting the U-Net architecture and its applications in sea ice monitoring and other domains. The fourth paragraph emphasises the combination of unmanned aerial vehicle (UAV) technology with deep learning techniques for efficient ice monitoring and categorisation; it also presents diverse models and their applications and the obstacles faced in the domain. The fifth paragraph highlights the importance of passive microwave data and deep learning in enhancing the prediction and monitoring of sea ice, underscoring the importance of this research field to the safety of maritime activities and climate studies.

Fundamental network topologies comprise convolutional layers that are interconnected in a manner that generates a pixel-wise prediction, which is then upsampled to produce segmentation maps. An example of such early architecture is FCN. The three base FCN models proposed by [Long et al. \(2015\)](#) are amongst the earliest deep networks that achieved notable performance in image segmentation. FCN substitutes the original fully connected layers in prominent networks, such as AlexNet ([Krizhevsky et al., 2017](#)), GoogleNet ([Szegedy et al., 2015](#)) and VGG

([Simonyan and Zisserman, 2014](#)), with a set of upsample decoders to produce pixel-wise segmentation outputs. The fully connected layers of each neuron are changed into deconvolution layers. FCN-based segmentation techniques have been implemented in numerous contexts, including instance-aware semantic segmentation ([Li et al., 2016](#)), brain tumour segmentation ([Mazumdar, 2019](#)), skin lesion segmentation ([Kaymak et al., 2020](#)) and iris segmentation ([Wang et al., 2019](#)). The utilisation of FCN has substantially advanced the performance of deep CNNs, although certain limitations, such as imprecise segmentation of small objects and need for complex training procedures, persist. Therefore, inspired by FCN's architecture, researchers have developed a sequence of improved deep networks designed for semantic segmentation tasks. SegNet was introduced by [Badrinarayanan et al. \(2017\)](#) as a symmetrical convolutional encoder–decoder framework to further enhance the FCN architecture. Convolutional layers are applied to generate image features during the encoding phase. During the decoding stage, an upsampling procedure is used to retrieve the feature map's detailed mapping. As a result, compared with FCN, SegNet uses less memory space and can be trained at a faster pace. In contrast to other architectures, SegNet possesses a notably reduced number of trainable parameters. On the basis of FCN, [Chen et al. \(2021\)](#) proposed an ensemble of SegNet-based models for segmenting rain-contaminated regions from X-Band marine radar images. To improve the FCN model, [Ronneberger et al. \(2015\)](#) introduced U-Net, which incorporates symmetrical encoder and decoder layers that are connected through skip connections to achieve excellent segmentation performance. The network works well on segmentation problems with numerous layers embedded into the model and considering different activation functions. In general, a layer is a collection of nodes or neurons that establish a mapping between the inputs and outputs of a network. A deep neural network consists of multiple layers that process input–output relationships that become progressively complex as the network depth increases. A novel approach known as PSPNet was introduced by [Zhao et al. \(2017\)](#), who suggested using CNN to parse the image and enhance performance. This method produces remarkable outcomes whilst balancing the demands of network complexity and efficiency. A notable advantage of these CNNs over threshold-based segmentation and traditional ML methods is that neither feature engineering nor threshold selection is required. Throughout the process of network training, convolutional layers autonomously extract feature and semantic information from the training data samples.

Expanding upon the remarkable efficacy of CNN in the context of sea ice concentration, many researchers have invested their time and effort to investigate deep model utilisation across other extensive datasets and domains. [Tamber et al. \(2022\)](#) utilised the binary cross-entropy loss function to train a CNN model and forecast the likelihood of ice formation. Furthermore, the authors applied augmented labels as a novel approach to improve the acquisition of information in the sea ice dataset. Multiscale MobileNet, which is also a deep learning framework developed by [Zhang et al. \(2022a\)](#), exhibits an average enhancement in classification accuracy of 4.86 % compared with the base SCNN model and 1.84 % performance improvement compared with the ResNet18 model. [Kortum et al. \(2022\)](#) integrated CNN with dense conditional random fields and adopted supplementary spatiotemporal background data to improve the resilience of the base model and accomplish accurate ice classification performance across multiple seasons.

Deep learning has achieved notable advancements in semantic segmentation across various domains in recent years. The U-Net network, in particular, has demonstrated excellent segmentation performance in various semantic segmentation tasks, so it has been extensively implemented in various applications. Moreover, researchers have investigated the use of the U-Net framework in determining the amount of sea ice. [Ji et al. \(2022\)](#) enhanced U-Net with an adaptive moment estimation optimiser and a batch normalisation layer and developed BAU-NeT, a dedicated model for sea ice application. Furthermore, in the study of [Gelís et al. \(2021\)](#), sea ice concentration prediction was performed by

utilising an improved FCN model developed based on the older version of the U-Net architecture. Meanwhile, Ren et al. (2020) introduced a U-Net-based model designed for the classification of SAR images of sea ice and open water. This model can precisely classify sea ice on a pixel-by-pixel basis. Afterwards, the authors conducted another study on the dual-attention mechanism, resulting in the development of a dual-attention U-Net model called DAU-Net. This model demonstrates enhanced segmentation accuracy compared with the U-Net model (Ren et al., 2023). Furthermore, Ren et al. (2022) introduced an enhanced iteration of the U-Net model that incorporates the dual attention mechanism to segment water and sea ice by using SAR images. Another research by Song et al. (2022) introduced the E-MPSPNet network, which is an architecture that integrates scale-wise attention with multiscale features. In contrast to widely used segmentation networks, such as HED-Net, DeepLabV3, PSPNet and U-Net, E-MPSPNet demonstrates commendable performance with a high level of efficiency.

In addition, Radhakrishnan et al. (2021) introduced an innovative training approach that utilizes curriculum learning with U-Net to enhance the stability of model training. The UNET++ framework, which was first introduced by Zhou et al. (2018), demonstrates excellent performance in tasks involving the segmentation of medical images. Then, Murashkin and Frost (2021) utilised UNET++ to map the Arctic sea ice in Sentinel-1 SAR scenes. According to the reviewed works, CNN has been effectively utilised not only in sea ice extraction tasks but also in the extraction of river and lake ice to enable the continuous monitoring of the evolution of glacial lakes on earth. This study also provides deep learning references for sea ice extraction work.

Ice monitoring has extensively used UAV technology because of its increased prevalence, reduced cost and high spatiotemporal resolution. Hence, data deficiency in satellite imagery can be partially remedied. In accordance with the CNN-based approach for ice monitoring, Cooke and Andrea Scott (2019) estimated sea ice concentration from SAR images by using the deep neural network DenseNet (Huang et al., 2017). Their deep learning approach has considerable potential in representing the internal content of data and achieving good performance in sea ice classification. The proposed deep learning approach considers robust representational features, thus allowing the effective extraction of diverse latent information embedded within the data. Wang et al. (2018) employed CNNs to classify sea ice in the Bering Sea by using SAR data. Song et al. (2021) utilised a combined model of CNN and long short-term memory to classify the sea ice in Hudson Bay by employing SI-STSAR-7 data. Meanwhile, Huang et al. (2021a) applied the U-Net deep learning model to classify Arctic sea ice by using SAR data. Another research by Kim et al. (2020) introduced a predictive model for monthly sea ice concentration that uses a basic CNN architecture. In addition, Khaleghian et al. (2021a) developed an enhanced version of the VGG-16 network to accurately separate ice and water in a binary form and classify different types of sea ice by using SAR data. The ICENET deep neural network architecture, which was designed by Zhang et al. (2020), contains an application that is closely related to the present work because it adopts semantic segmentation CNN to detect river ice in aerial drone footage of rivers. ICENET has high efficacy in differentiating open water, surface ice and background classes. However, the majority of current CNN-based approaches concentrate on analysing SAR sea ice images and river-ice data (Wang et al., 2016).

Kim et al. (2019) classified various ice objects by utilising a basic CNN model. They concentrated on ice objects, such as large ice floes and icebergs, as opposed to classifying ice types. Han et al. (2021) constructed a CNN-based architecture to classify sea ice on the basis of optical and SAR images. The results showed that the integration of diverse features from multiple sources of image data results in a notable enhancement. Moreover, Boulze et al. (2020) suggested utilising sentinel-1 SAR data in neural networks to classify sea ice. In this approach, satellite imagery is needed instead of onboard camera footage to classify sea ice. Another comprehensive deep model assessment was performed by Singh et al. (2020), who measured the effectiveness of

various deep neural network models in semantic segmentation of river ice images. Meanwhile, Dowden et al. (2021) examined two semantic segmentation frameworks, namely, PSPNet and SegNet, for segmenting and detecting sea ice in images captured by onboard cameras. The segmentation experiment revealed that both approaches produce inaccurate segmentation outcomes. Alsharay et al. (2022) devised a raindrop removal method to enhance classification performance under the influence of droplets on the segmentation outcomes of captured images. A semantic segmentation model that uses a conditional generative adversarial network was also introduced in reference to Alsharay et al. (2023). This model exhibits commendable robustness and effectively mitigates the effect of rainfall on the segmentation outcomes. In addition, Chen et al. (2022b) proposed a multi-label model integrated with squeeze and excitation modules and used this model with airborne images to classify sea ice. The model demonstrates a substantial accuracy enhancement over standard machine learning techniques, such as RF and gradient boosting decision tree.

Deep learning methods have also been applied to predict sea ice concentration on the basis of daily data collected by passive microwave sensors, including SMMR, SSM/I and SSMI/S (Andersson et al., 2021; Zheng et al., 2022). The research of Gao et al. (2019) achieved notable advancements in detecting and tracking sea ice variations from SAR images by utilising collaborative representation to produce a transferred multilevel fusion network, which serves this purpose excellently. This research is crucial in guaranteeing the security of maritime operations and streamlining the extraction of natural resources. Moreover, Chen et al. (2023) used passive microwave data and reanalysed the data to accurately forecast sea ice concentration. This step not only ensures safe navigation in Arctic regions but also offers valuable historical information for investigating climate changes in such regions.

In summary, the implementation of deep learning to the segmentation of sea ice regions has specific limitations. A major drawback of the system is its need for a large amount of labelled sea ice data for model training purposes. This problem is common in various applications, where the deep learning model faces difficulty in achieving optimal fitting (Min et al., 2022). However, at present, extensive and inclusive reference datasets that accurately depict the whole scenario are unavailable. Table 4 shows an overview of the deep learning approach for sea ice concentration mapping. Fig. 5 presents the segmentation results of a sample SAR image with the respective indicator details.

7. General discussion

Prolonged changes in polar regions considerably affect the balance between sea ice and the polar ocean system, which will disrupt the system's capacity to moderate the climate, precipitation and atmosphere. Moreover, collapsing ice shelves introduce additional complexity and uncertainties to the ocean-sea ice-atmosphere system that disrupt the optimal interactions between sea ice and polar seas. The continuous reduction in sea ice areas in polar regions and the increased erosion rate caused by glaciers and river flows are expected to generate changes in the environment, especially in terms of the rate of absorption and desorption of bioactive gases. Therefore, an accurate sea ice thickness mapping system is necessary to comprehend and forecast the consequences of the earth's changing climate, specifically on the operational activities in ice-covered waters. Satellite-based monitoring systems of sea ice thickness must be promoted to conduct continuous measurements. The monitoring process needs to use a continuous time series, which is a crucial data input in ascertaining whether recent observations stem from an interannual variability that is inherent in the polar climate or constitute a sustained negative trend.

SAR-based sea ice concentration monitoring systems have shown promising performance because of the exceptional level of detail captured by this modality. SAR technology allows the precise identification of sea ice, ice types and other important characteristics that are relevant to ship navigation in ice-covered areas. RADARSAT and ERS/

Table 4

An overview of the deep learning approach for sea ice concentration mapping.

Reference	Year	Dataset	Application	Algorithm	Advantages	Disadvantages
Yan and Huang (2018)	2018	GNSS-R	Sea ice detection and sea ice concentration prediction	• Delay-Doppler maps (DDM) • CNN	Evaluates the performance of the CNN using both full-size and cropped DDM images.	Findings may not be universally applicable across all conditions or datasets, as the CNN's performance was evaluated under specific scenarios.
(Cooke and Andrea Scott, 2019)	2019	Passive microwave data	Estimated sea ice concentration	DenseNet	Achieves competitive accuracy with passive microwave data	CNN did not accurately capture several distinct ice characteristics that were not adequately included in the training dataset.
(Kim et al., 2019)	2019	Close-range optical imagery of ice conditions	Classifying ice type	ResNet18	Model presented in the paper can be used for accurate multilabel ice object classification from close-range optical imagery	The model has lesser performance on minority classes compared to majority classes, and its effectiveness is restricted to clear images.
(Ren et al., 2020)	2020	SAR images	Classifying sea ice and open water pixels	U-Net	Automate the classification process using the proposed deep learning model, reducing the need for manual labelling and increasing efficiency	Only focuses on SAR imagery captured in the Gulf of Alaska near Bering Strait, limiting the applicability of the proposed model to other regions
(Zhang et al., 2020)	2020	NWPU_YRCC dataset	River ice Segmentation	ICENET	Use of unmanned aerial vehicle (UAV) aerial photography in ice monitoring provides higher spatial and temporal resolution	Dataset is still not large enough to cover all realistic scenes
(Singh et al., 2020)	2020	Digital images and videos	Semantic segmentation of river ice images	• SVM • DeepLab • U-Net • DenseNet • SegNet	Effectively handle the task of distinguishing between two types of river ice from digital images	Limited availability of labelled training data
Zhu et al. (2020b)	2020	GNSS-R	Sea ice monitoring	• Decision tree • Random Forest (RF)	The study shows that using feature sets rather than single observable methods gives an improved understanding of the reflecting surface properties.	GNSS-R data used in the study is not available every day, which can lead to gaps in the monitoring process.
(Han et al., 2020)	2020	Remote sensing image	Classifying remote sensing sea ice images	• 3D-CNN • Squeeze-and-excitation (SE) networks • Support vector machines (svms) • DeepLab v3	This study improves the accuracy of classifying sea ice images in remote sensing by considering the connection between spectral features and addressing the issue of limited sample size.	Limited to three specific data sets, which may not fully represent the diversity of images of sea ice obtained by remote sensing.
(Tom et al., 2020)	2020	Lake Ice Detection	Data from two winters (2016–17 and 2017–18) and three alpine lakes in Switzerland	• DeepLab v3	Demonstrates good generalization across unseen lakes and winters through cross-validation tests	Insufficiently dependable reference data during the period of changing dates and the existence of clouds and foggy webcam images in Sentinel-2 data
(Dowden et al., 2021)	2020	Sea ice classification	Nathaniel B. Palmer dataset	• SegNet • PSPNet101	Enabling real-time sea ice classification and monitoring during Antarctic expeditions	Evaluation is based on a relatively small dataset of 240 labelled images, which may not fully capture the variability and complexity of real-world sea ice conditions
(Radhakrishnan et al., 2021)	2021	SAR images and passive microwave data	Estimation of sea ice concentration	U-Net	Improve estimations in marginal ice zones while preserving accurate figures in consolidated ice.	Evaluation charts based on image analysis are less precise, contingent on analyst interpretation, and restricted to particular geographic regions.
(Murashkin and Frost, 2021)	2021	Sentinel-1 SAR images	Automated sea ice mapping	UNET++	Addresses the border effect that appears when a tiled classification is applied	Paper does not explicitly mention the limitations
(Huang et al., 2021a)	2021	SAR images	Classify Arctic Sea ice	U-Net	Enables an “end-to-end” classification process and reduces the time and effort required for sea ice classification tasks	Only focuses on classifying three specific sea ice types
(Khaleghian et al., 2021b)	2021	SAR images	Separate ice and water in binary form	VGG-19	Models produce accurate classification results both qualitatively and quantitatively	The presence of additive system noise in SAR data poses a significant challenge in obtaining accurate and detailed sea ice maps.
(Han et al., 2021)	2021	SAR images and optical data	Classify different types of sea ice Sea ice image classification	• Improved SPP • Path aggregation network (panet)	Demonstrate the effectiveness of the proposed method, with classification accuracies of 96.61 % and 95.69 %.	Does not provide any insights into the scalability or applicability of the proposed method
(Ansari et al., 2021)	2021	Detection and characterization of river ice	Collected data from Dauphin River, in Manitoba, Canada	Icemasknet	The fast and accurate data acquisition provided by icemasknet can contribute to	Only focuses on river ice detection and characterization using aerial imagery, without considering

(continued on next page)

Table 4 (continued)

Reference	Year	Dataset	Application	Algorithm	Advantages	Disadvantages
(Ji et al., 2022)	2022	Remote sensing satellite optical images	Identifying different components of Arctic Sea ice during summer	BAU-NET	improved understanding of river characteristics Enhances the time efficiency and accuracy of sea ice predict in comparison to alternative approaches, achieving a remarkable accuracy rate of 97.26 %.	other remote sensing techniques or data sources Not mentioning the exact dataset or satellite images used for testing and training
(Ren et al., 2022)	2022	Sentinel-1A SAR images	Segment water and sea ice	DAU-Net	Optimizing inaccurate features and boosting feature discriminability A more compact model and quicker processing speed	Require further SAR images to assess the suitability of the model.
(Song et al., 2022)	2022	AI4Arctic/ASIP	Ice-water segmentation	E-mpspnet	Possess a significant level of precision in identifying ice formation on wind turbines	Effectiveness of the segmentation is constrained by the quality and availability of ice charts Working with little data
(Hacrefendioğlu et al., 2022)	2022	Photos of wind turbines taken under different conditions	Ice detection on wind turbine blades	- Pretrained model: ResNet-50, VGG-16, VGG-19, Inception-V3 - Visualization techniques: Grad-CAM, Grad-CAM++, and Score-CAM		
(Alsharay et al., 2022)	2022	Nathaniel B. Palmer dataset	Sea ice identification	- VGG-16, - FCN - PSPNet	Reducing the need for large labelled datasets and saving computational resources	Does not compare the proposed raindrop removing framework with other existing methods or techniques for raindrop removal in the context of sea-ice identification
Yang et al. (2023)	2023	Sea ice forecasting	GNSS-R	- Delay-Doppler maps (DDM) - Deep Neural Network (DNN)	The work establishes a correlation between delay-Doppler maps (DDMs) and sea ice concentration through the application of deep learning and statistical models. DAU-Net model's ability to deal with complex sea conditions and its good robustness	The TDS-1, which is the latest technology, is unable to capture daily changes in sea ice concentration dynamics.
(Ren et al., 2023)	2023	Sentinel-1A SAR images	Classifying sea ice	DAU-Net		Misclassifications may be due to the lack of training samples

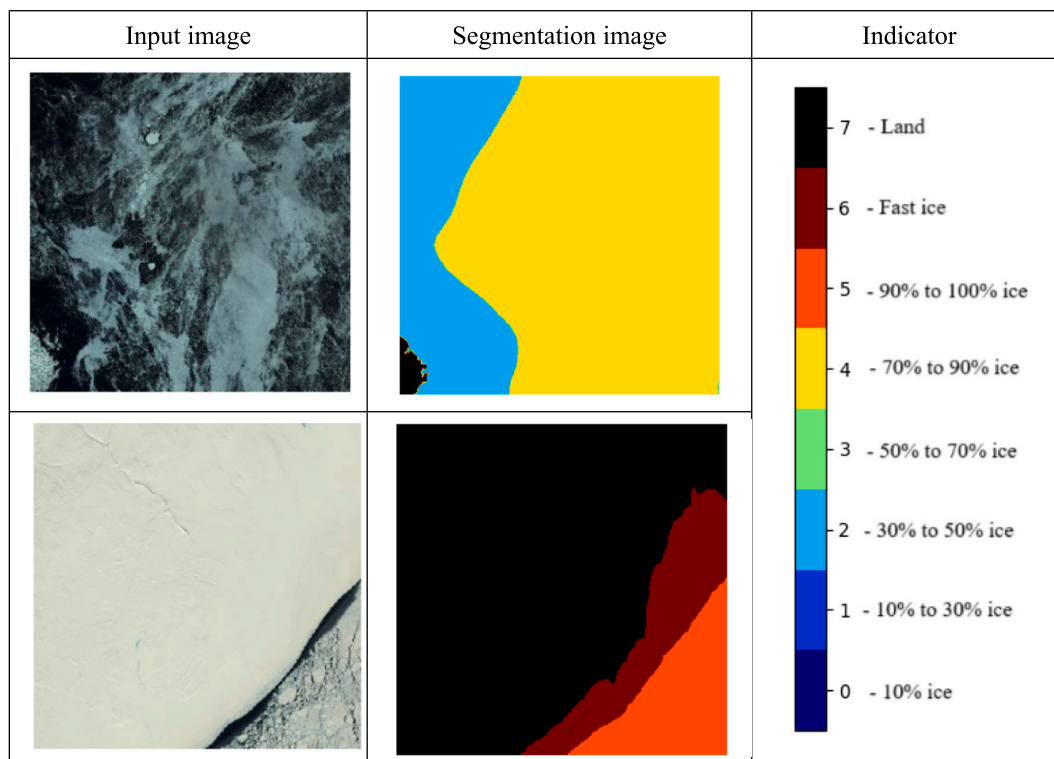


Fig. 5. Samples of input images and their respective segmentation results.

ENVISAT programmes play a crucial role in advancing SAR-based ice surveillance initiatives that focus on aiding navigation in ice-covered regions. Since 2014, the Sentinel-1 satellite, which provides SAR data, has been consistently used to monitor maritime lanes in the Arctic. The SAR images of polar regions consistently produced by this satellite offer vital insights into the dynamics of sea ice, including ice motions, leads, ridges, rough ice patches, rapid ice and massive ice formations. Moreover, the satellite's high-quality imaging allows navigators to discern and categorise various forms of Arctic sea ice that can contribute to the comprehension of sea ice formation, motions and melting patterns, amongst other dynamics. Thus, accurate climate models can be developed to forecast any abrupt change in the sea ice balance.

Many researchers have focused on developing dedicated deep learning systems to categorise sea ice by using high-resolution SAR data, and these systems have been practically applied to monitor and measure sea ice data. Different from lengthy manual interpretation conducted by professionals, deep learning-based models offer swifter and more effective techniques for producing high-resolution maps of sea ice that include its concentration and type. These automated methods enable the continual surveillance of sea ice conditions, hence providing valuable insights to various stakeholders. Specifically, they aid in comprehending the earth's global energy equilibrium, the thermodynamic interactions between the ocean and atmosphere, the weather patterns in mid-latitudes and the influence of climate change on polar regions. The practical ramifications of this sea ice study transcend multiple domains, specifically in maritime logistics, where accurate and instant data regarding the concentration and nature of sea ice are invaluable to ensure safe and secure navigation.

7.1. Challenges

Segmentation for sea ice mapping presents several challenges and concerns. Firstly, the availability of open data sources is severely constrained, thereby compromising the use of AI models to learn accurate sea ice concentration, especially in cases involving high-resolution data. The limited availability and fragmented nature of current data sources pose challenges in fully capturing sea ice characteristics. Secondly, existing methods for segmenting sea ice exhibit low mapping performance when confronted with intricate sea ice situations. Normally, the sampled sea ice data consist of a wide range of variations in terms of shape, compactness, depth and other features, which poses a critical challenge to conventional algorithms. Thirdly, numerous complicated changes, such as fissures, ridges and leads, occur during sea ice formations; they are difficult to capture and depict accurately, especially when non-deep learning techniques are used. Last, the presence of submerged ice poses difficulty to remote sensing-based approaches, whose sensors are unsuitable for acquiring structure and dimension information for these sea ice cases. Additional investigation of hybrid sensing is required to address this issue and realise improved identification of sea ice through enhanced modelling and mapping strategies.

Utilising SAR data for navigation is challenging because of the limited sampling frequency and slow data transmission in Arctic regions. Typical transmitted data have a size of 200–300 megabytes, and SAR images will take hours to complete this transmission via Iridium. Therefore, transmitting multiple images daily in Arctic regions by using current communication services is impossible. In the next few years, the availability of satellite broadband services will possibly alleviate this difficulty. Projections show that the shipping industry in Arctic regions will expand substantially in the near future as a result of increasing tourism activities, the extraction of oil and gas by Russia and the desire of numerous nations to investigate potential valuable minerals and other reserves. Various emerging technologies have been proposed to resolve these challenges. Satellites and UAVs are being progressively outfitted with onboard AI-driven data processing capabilities, enabling initial processing at satellites or nearby ground stations, to alleviate the strain on communication networks by broadcasting only the most pertinent

information. Moreover, deep learning-based compression algorithms, such as generative adversarial networks (GANs), can provide essential image characteristics whilst substantially decreasing file sizes, which can enhance the efficiency of real-time SAR data transfer for efficient updates. The implementation of low-earth-orbit satellite constellations (e.g. Starlink and OneWeb) offers enhanced bandwidth and reduced latency, hence advancing Arctic communication and facilitating frequent SAR image transmission for near-real-time sea ice monitoring. Furthermore, AI-driven analytics pipelines allow for the immediate categorisation and segmentation of sea ice prior to transmission, thus improving deep learning applications in Arctic operations and guaranteeing prompt updates for safe navigation and effective monitoring processes.

Deep learning applications for sea ice concentration monitoring by utilising SAR data encounter numerous challenges. An example of such a challenge is speckle noise, which is an intrinsic attribute of SAR images resulting from the coherent nature of radar data processing. This noise presents granular patterns that hide intricate details, hindering deep learning models' ability to extract features reliably. Preprocessing techniques, such as adaptive filtering, non-local means filtering and deep learning-based denoising methods, have been extensively employed to address this issue. Specifically, models derived from GANs and CNNs have been utilised for speckle reduction to enhance segmentation precision without compromising image resolution. Another critical challenge is the varying contrast of SAR images, which is affected by environmental factors, such as surface roughness, wind conditions and sensor incidence angle. These differences influence categorisation accuracy by modifying pixel intensity distributions. To resolve this issue, researchers apply contrast normalisation methods, including histogram equalisation and deep learning-based contrast enhancement, to calibrate brightness discrepancies across images and enhance segmentation accuracy. Attention-based deep learning models have also been developed to dynamically modify feature importance so that enhanced sea ice segmentation outputs can be generated.

Aside from image quality issues, the restricted data transmission velocity and communication limitations in the Arctic pose serious obstacles to real-time SAR data transfer. This constraint diminishes the system's capability to deliver prompt updates for navigation and monitoring purposes. Real-time data processing at the edge, which can address this issue, has elicited much attention. Preliminary data processing at satellites or nearby ground stations allows for the transmission of only the most pertinent information, thus alleviating the strain on communication networks and facilitating near-real-time decision-making. These techniques enable deep learning models to be applied effectively to manage the distinct attributes of SAR data, enhance sea ice segmentation performance and improve classification efficacy for sea ice monitoring activities.

Another critical barrier in utilising deep learning methods for sea ice monitoring is the scarcity of comprehensive, high-quality datasets that encompass various circumstances. Many existing datasets are region- or season-specific, which may reduce the generalisability of deep models when tested across various polar conditions. Another issue is data sparsity in specific places and the absence of ground truth data for validation purposes. Inconsistencies in the manual annotation for producing ground truth labels might also inject noises into the training process, thus affecting deep model accuracy. Given these challenges, various initiatives have been introduced by researchers to provide standardised annotated datasets that include synthetic data augmentation to enhance data quality and coverage and create precise AI models for sea ice monitoring.

7.2. Opportunities

With advancements in sea ice segmentation and classification modelling, researchers need to extract good spatial distribution and dynamic variation information from sea ice regions. A wide range of

relevant geographic information has been produced and shared by various studies through meticulous investigation and review processes. These findings have a crucial effect on multiple fields, such as weather prediction (Rogers et al., 2013), marine security (Chang et al., 2015) and environmental preservation (Chen et al., 2019). In this section, several research opportunities in specific application domains are explored.

1. Weather prediction

The outputs of sea ice mapping exert a considerable effect on meteorological prediction and climate analysis. By employing remote sensing to map sea ice information, these outputs allow predictive models to provide insights into the climate changes that occurred between the atmosphere and the ocean, thereby consolidating our comprehension of sea ice dynamics (Mäkynen et al., 2020). In addition, the importance of monitoring sea ice conditions cannot be downplayed because of its role in balancing the ocean's energy through its albedo properties (Jäkel et al., 2019). Furthermore, the limited thermal conductivity of sea ice has a substantial effect on the heat transfer that occurs between the atmosphere and the ocean. Therefore, sea ice condition is an essential barometer to detect any change in the world's climate. By conducting routine sea ice monitoring through extraction and classification functions, we can 1) track the temporal and spatial fluctuations of sea ice, 2) examine the patterns of sea ice formation and retreat and 3) furnish empirical evidence to support investigations on climate change.

Therefore, intelligent analysis using segmentation and classification outputs is crucial for climate change monitoring, climate prediction and meteorological forecasting. Through the application of remote sensing techniques, scientists can extract and categorise various critical data by using sea ice statistics. This information allows scientists to improve the forecasting capability of climate change models, which facilitates a comprehensive understanding that links sea ice and climate conditions and thus enables the automated evaluation and tracking of climate change effects.

2. Deep learning and climate change monitoring

The deep learning approach has substantially enhanced climate change monitoring by facilitating an accurate, scalable detection of sea ice concentration patterns. The use of AI-driven models on high-resolution imaging datasets enables researchers to monitor fluctuations in sea ice extent, concentration and melting trends over time. These advancements support climate change research by monitoring long-term sea ice trends; deep learning models trained on historical and real-time sea ice data can forecast seasonal and decadal variations in Arctic and Antarctic ice coverage. The generated outputs are essential for comprehending climate variability and evaluating the effects of global warming. Moreover, AI-driven segmentation and classification of sea ice components in SAR and optical images can enhance climate simulation models, hence increasing the models' accuracy in climate change prediction. Dependable AI-based sea ice monitoring systems can also contribute to the development of improved climate adaptation strategies, marine conservation initiatives and policymaking by delivering real-time secondary data on ice conditions, which are crucial for addressing the effects of climate change on polar ecosystems and maritime activities. Integrating deep learning into climate change research enables scientists to obtain profound insights into the relationships amongst sea ice, ocean circulation and atmospheric changes, underscoring the importance of AI in climate change adaptation and mitigation efforts.

3. Marine security

Precise extraction and categorisation of sea ice statistics are crucial for ocean and maritime navigations. Valuable insights, including those

regarding route planning tools, ice edge charts and maps of sea ice distribution, can be generated in an efficient manner by employing remote sensing methodologies in the extraction and categorisation of sea ice data. These insights provide indispensable aid to vessels, facilitating their safe navigation by circumventing ice-prone regions.

The navigational implications of using sea ice extent and thickness information are substantial, as emphasised by Buixadé Farré et al. (2014). Multi-year ice (MYI), which is notorious for its extreme rigidity and thickness, presents serious hazards to vessels. On the contrary, sea ice, which is younger and thinner than MYI, can be dealt with safely by using icebreakers to help conventional cargo vessels traverse ice-free coastal regions with great ease during summer (Zhou et al., 2021). Furthermore, Chen et al. (2022a,c) demonstrated that the thickness of sea ice exerts a more substantial influence on the velocity of vessels than ice concentration does, emphasising the importance of the former in facilitating effective passage along the Arctic route. The consequences of these discoveries are crucial for Arctic shipping and sea ice prediction, which facilitates safe navigation and efficient decision-making throughout sea ice areas.

4. Environmental preservation

Deep learning methods are essential for assessing the effects of climate change on sea ice through the accurate and precise evaluation of extensive remote sensing data. Through automated and intelligent monitoring processes, deep learning methods reduce the need for manual analysis, facilitating the continuous and real-time surveillance of polar regions. Furthermore, these intelligent models aid in comprehending and alleviating the extensive ramifications of climate change, including its effects on marine ecosystems, coastal populations and global sea levels. Coastal natural resource development is highly dependent on statistics related to sea ice. Intelligent methods of extracting and categorising sea ice can facilitate rapid evaluation of its effects on various operations, including submarine cable installation (Taormina et al., 2018), oil and gas extraction (Desmond et al., 2019) and fishing (Hunt et al., 2021). This information provides decision-makers with crucial benchmarks to economically implement commercial activities.

Moreover, sea ice is a vital component of the polar ecosystem. The process of freezing and melting of sea ice not only regulates temperature fluctuations in polar regions but also influences the stability of oceanic stratification, salinity and temperature, thus exerting an influence on the circulation of the global ocean (Li and Liu, 2022). The process of extracting and categorising sea ice can yield valuable statistics, including those on the delineation of sea ice boundaries, identification of ice–water interfaces and detection of cracks. This information is crucial in ecological research and conservation endeavours. The findings of AI-based analysis can be applied to diverse domains of marine science (Sreejith et al., 2022; Yan et al., 2020), such as marine geology, marine biology and marine physics.

7.3. Innovative contributions

This review presents a distinct perspective by highlighting issues specific to SAR data and providing innovative AI-driven solutions, in contrast to early studies that focused on conventional methodologies and basic remote sensing approaches. The integration of modern denoising techniques, attention-based models and multi-modal data fusion methods is examined to improve the accuracy and efficiency of sea ice concentration monitoring. Hence, this study has numerous distinctive features that set it apart from previous studies in the domain of sea ice concentration monitoring.

1. Targeted assessment of AI methodologies on SAR data

This review analyses deep learning applications for SAR data and

emphasises certain challenges, such as speckle noise and contrast fluctuation. Generally, prior reviews only examined basic sea ice mapping techniques. By contrast, this review presents preprocessing and model-specific advances, including deep learning-based denoising techniques and contrast normalisation algorithms, for addressing the aforementioned challenges.

2. Proposed framework for the integration of multi-sensor data

This review presents a multi-modal data fusion architecture that combines SAR, optical and passive microwave data. The presented architecture augments the AI model's generalisability and performance under various conditions. It mitigates the constraints caused by limited datasets as acknowledged in prior studies, thus providing a good framework for subsequent research to address data scarcity.

3. Prospective research opportunities in using the AI approach

This review explores novel prospects for AI applications in climate change surveillance, including predictive modelling and real-time sea ice monitoring. This review also highlights the usage of transfer learning and synthetic data augmentation to create a framework for constructing scalable and adaptive models for worldwide applications.

7.4. Comparison with similar studies and broader implications

Several recent studies have investigated deep learning techniques for sea ice concentration monitoring, mainly utilising Sentinel-1 SAR data. Huang et al. (2023) developed DF-UHRNet, a CNN model that integrates high- and low-level fusion modules with attention mechanisms to classify sea ice, including thin ice, one-year ice, open ocean, and multi-year ice. Their research exhibited improved pixel-level precision, especially in differentiating ice-water interfaces. Similarly, Pires de Lima et al., (2023) introduced a deep learning architecture that incorporates atrous convolutions into neural networks, merging ResNets with Atrous Spatial Pyramid Pooling to enhance the robustness of sea ice segmentation, especially under complex conditions consisting of noise and diverse ice textures.

Compared to these studies, our review highlights multi-source high-resolution imaging datasets that integrate SAR, optical, and drone-based imagery for monitoring sea ice concentration. While earlier studies predominantly concentrate on SAR-based classification, our research thoroughly evaluates deep learning models across multiple imaging modalities, including data fusion and scalability issues. Moreover, our research examines benchmarking methodologies utilising Canadian Ice Service (CIS) ice charts, ensuring comparable evaluation metrics across various AI models.

Our findings correspond with the increasing trend of utilising deep learning models for sea ice segmentation and classification. Our study, however, extends SAR-based approaches by integrating high-resolution optical and drone datasets. This integration improves spatial resolution and feature extraction, facilitating a more accurate sea ice concentration estimation. Furthermore, our research emphasises the necessity for a standardised benchmarking approach to ensure consistent evaluation of AI models across datasets. Our findings highlight the significance of high-resolution imaging datasets in improving real-time navigation safety within maritime surveillance systems. In contrast to scatterometry or passive microwave sensors that yield coarse-resolution data, SAR and optical images facilitate fine-scale segmentation essential for Arctic operations. Our study highlights the importance of AI-driven feature extraction methods, especially in addressing the problems presented by variations in sea ice concentration.

8. Conclusion

This review highlights the transformational potential of AI,

particularly deep learning methods, in the automated monitoring of sea ice concentration, which is a crucial indicator to detect climate change. The application of AI has improved the precision of sea ice mapping and provided insights into the Arctic environment; these insights, in turn, can be used to facilitate informed decision-making for environmental conservation, maritime safety and resource management. This review presents a comprehensive summary of the techniques employed for sea ice mapping and includes the respective techniques' strengths and weaknesses. The summarization is arranged into three main categories: conventional, machine learning and deep learning approaches. Furthermore, this review offers a curated compilation of accessible open-source datasets for sea ice classification and segmentation, the two most important components of high-level analysis of the sea ice environment. This work emphasises the transformative potential of AI technology in enhancing sea ice concentration monitoring via high-resolution imaging datasets. The reviewed datasets are mainly from SAR, optical and drone sensors that provide high-resolution imaging, which is crucial in improving the precision and accuracy of sea ice concentration mapping by addressing the issues of speckle noise, resolution constraints and data sparsity. Although other sensor modalities, such as scatterometry, altimetry and passive microwave sensors, can also yield useful data for extensive monitoring of sea ice concentration, their integration with AI is not the scope of this review, which is a promising domain for further study. This review concentrates on high-resolution imaging datasets, offering a focused perspective on recent achievements whilst emphasising the potential future research directions in sea ice concentration monitoring. Future studies could build upon the findings discussed in this paper by exploring multi-modal data integration and extending AI applications to other remote sensing modalities to further enhance the accuracy and scalability of sea ice concentration monitoring systems. This work also discusses several possible opportunities for future research on cover detection techniques, model methodologies and cartographic applications.

CRediT authorship contribution statement

Marzuraikah Mohd Stofa: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Siti Raihanah Abdani:** Writing – review & editing. **Asraf Mohamed Moubark:** Writing – review & editing. **Muhammad Ammirul Atiqi Mohd Zainuri:** Writing – review & editing. **Ahmad Asrul Ibrahim:** Writing – review & editing. **Nor Azwan Mohamed Kamari:** Writing – review & editing. **Mohd Asyraf Zulkifley:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Mohd Asyraf Zulkifley reports financial support was provided by Malaysia Ministry of Higher Education. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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